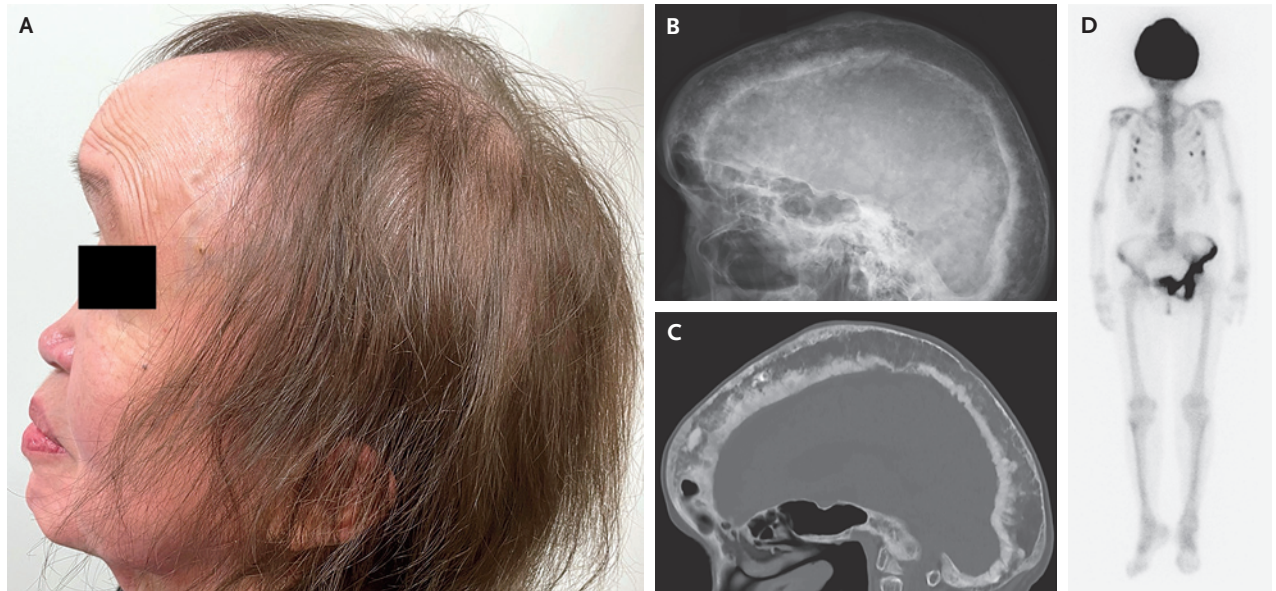


IMAGES IN CLINICAL MEDICINE

Paget's Disease of Bone



AN 80-YEAR-OLD WOMAN PRESENTED TO THE ENDOCRINOLOGY CLINIC with a several-year history of gradual forehead enlargement and progressive hearing loss. She reported no headache or bone pain. Sixteen years earlier, mild thickening of the skull had been noted on imaging studies that had been conducted to evaluate early-stage breast cancer, which was successfully treated. However, the patient had not followed up on the incidental radiographic finding as recommended. At the current presentation, physical examination was notable for frontal bossing (Panel A). The alkaline phosphatase level was 592 IU per liter (reference range, 38 to 113). A radiograph of the skull showed marked thickening of the calvarium with irregular patches of sclerosis and lucency, resulting in a cotton-wool appearance (Panel B). Whole-body computed tomography showed no evidence of cancer but did reveal severe cortical thickening, sclerosis, and focal osteolysis of the skull (Panel C). Audiometry confirmed sensorineural hearing loss. A radionuclide bone scan revealed increased uptake in the skull and the left side of the pelvis (Panel D). A diagnosis of Paget's disease of bone was made. Treatment with a bisphosphonate was initiated. At a 3-month follow-up visit, the alkaline phosphatase level had decreased, but the hearing loss and frontal bossing remained unchanged.

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